Multiple Magnification Learning: A Deep Learning Approach for Classifying Breast Tumor from Histopathological Images based on the Concept of Multiple Instance Learning

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Abstract. The classification of breast cancer tumors in histopathological images is an expanding research field in computer-aided diagnosis. While the conventional approach to breast tumor classification involves extensive labor in hand-engineering features used for classification, the development of deep learning techniques has enabled a new approach that requires less human involvement in feature extraction. The Convolutional Neural Network (CNN) is one of the deep learning techniques that allows for the learning of feature representation given only input images. The development of CNNs has been common in many computer vision problems and is recently employed in histopathological image analysis. This research proposes an approach of Multiple Magnification Learning (MML) that utilizes the concept of Multiple Instance Learning (MIL). It employs a deep Convolutional Neural Network (CNN) model with four input paths, which simultaneously processes images at four different magnification levels. The backbone network used in the model is EfficientNetV2-S for histopathological image classification. The proposed method significantly outperforms previous state-of-the-art approaches in terms of accuracy, precision, recall, F-measure, Matthew's Correlation Coefficient (MCC), and Area Under the Curve (AUC) when evaluated on an independent test dataset.

Keywords: breast cancer · histopathological image · BreakHis · Convolutional Neural Network · Multiple Instance Learning.

1 Introduction

Breast cancer stands as a prominent cause of death among women worldwide. It is revealed that it contributes to nearly half a million fatalities globally,

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with an annual incidence of 1.7 million new cases [4]. Timely detection plays a pivotal role in achieving a favorable prognosis in breast cancer treatment [4, 28]. Among various detection methods, histopathology remains the gold standard for precise identification [28]. This involves examination of suspected and stained tissue lesions under a microscope at varying magnifications [26]. However, histopathology's reliance on expertise from histopathologists renders the process time-consuming and burdensome. Hence, there arises a crucial need for the development of an efficient and automated tool capable of delivering accurate and prompt diagnoses of histopathological images.

Presently, Artificial Intelligence (AI), particularly Deep Learning, has gained significant public attention due to its potential across various applications. Deep Learning has been extensively applied in diagnostic tasks, such as tumor segmentation [17], classification of tumor types using ultrasound images [15], and the identification of lung cancer in X-ray images [19]. These endeavors have yielded promising results, reaffirming the efficacy of data-driven methods in diagnosis. In Breast Cancer recognition, Deep Learning applications have long been studied. Numerous research has employed Deep Learning techniques to identify Breast Cancer in histopathological images [8, 27, 9]. Many of these works have centered on the BreakHist dataset [1], a widely used and publicly available dataset containing histopathological samples of both benign and malignant breast tumors at varying magnification levels (ranging from 40x to 400x). Despite the encouraging outcomes from existing approaches, it is noteworthy that most models focus on a singular magnification. On the other hand, in clinical practice, histopathological samples usually undergo scanning and digitization at different magnification levels as pathologists may switch between different levels to recognize distinct morphological features to produce accurate results [25, 18]. Given this perspective, the classification of tumor types relying on multiple magnification levels becomes imperative to align with established clinical practices.

One of the recent efforts to identify breast tumor types is the Multiple Instance Learning (MIL) method [22]. In MIL, an image is split into patches (instances), the model then considers a "bag" of these patches of the same sample. Nevertheless, the misclassification of a single instance as positive has the potential to result in the mislabeling of the entire sample or 'bag', thereby hindering its effectiveness in practice. On the other hand, while MIL possesses the capacity to make decisions based on multiple instances, it exclusively treats instances as patches from the same image, thus also restricting its applicability to a specific magnification level only. To tackle these challenges, we introduce a novel approach to breast tumor classification in this paper, termed Multiple Magnification Learning (MML) inspired from MIL. Our approach is designed to offer both efficient and accurate predictions of breast tumor lesions while addressing the aforementioned limitations of MIL. The contributions of this paper are twofold:

- We introduce a novel approach, Multiple Magnification Learning, capable of classifying breast tumors using instances from various magnification levels. In contrast to MIL, our method avoids excessive reliance on a singular magnification level or a lone instance.

– Through experimentation on the BreakHis dataset, we demonstrate that our approach outperforms existing baselines by a considerable margin, affirming its potential for practical applications.

2 Related Works

Till now, Deep learning has been extensively investigated for breast cancer pathological image classification. In 2018, Erkan Deniz proposed the usage of pretrained VGG16 [12] and AlexNet [11] for feature extraction, and employed a Support Vector Machine classifier to recognize Breast Cancer based on histopathological images [8]. In 2020, Masheh Gour published a research work and introduced a novel Convolutional Neural Network (CNN) architecture named ResHist for classification of breast cancer histopathological images [9]. In 2021, a study suggested employing DenseNet121 for the classification of breast cancer histopathological images across four distinct magnification levels [27].

From a technical standpoint, these Convolutional Neural Network (CNN)-based methodologies require resizing input images to a predetermined resolution, potentially resulting in the loss of discriminative information. To address this constraint, the adoption of Multiple Instance Learning (MIL) is introduced [22]. In MIL, authors extract 64x64 patches from a sample, treating these patches as a collective "bag" of instances. Classification decisions are subsequently made based on the collective characteristics of this bag of instances, diverging from the conventional practice of relying on a singular resized image, as seen in earlier approaches.

Despite showing premises, we recognized that these existing research focus on magnification-specific methodologies, prompting a consideration of their alignment with clinical practices where pathologists routinely assess samples at various magnification levels during examination [25, 18]. Consequently, there arises a necessity for the development of a specialized Deep Learning method capable of making predictions using data from different magnification levels. This, in turn, calls for the modification or enhancement of existing architectures to accommodate multiple inputs simultaneously.

In this context, MIL emerges as the closest and most pertinent solution with the capability to address such a requirement. However, it is crucial to acknowledge certain limitations associated with the use of MIL: (a) MIL inherently treats each image patch as an instance [22], overlooking the potential to treat each magnification level image as an instance; and (b) The misclassification of an instance in MIL can significantly impact the final results [22, 2]. These considerations underscore the need for further refinement and exploration in the development of Deep Learning methods tailored to handle data of diverse magnification levels in pathological image classification tasks. To this end, we proposed our approach named Multiple Magnification Learning. Our method is inspired from the original MIL work [22], while concurrently addressing its inherent limitations. Our methodology demonstrates the capability to effectively handle inputs from various magnification levels, mitigating the tendency to rely excessively on a singular

instance. The specifics of our method are comprehensively described in Section 3.

3 Methodology

In this study, Multiple Maginification Learning (MML) is employed based on the concept of Multiple Instance Learning (MIL) [5], where a "bag" consists of four images captured at different magnification levels. In conventional Multiple Instance Learning (MIL) approaches [5], the misclassification of a single instance as positive could lead to the entire collection of instances, or 'bag', being mislabeled similarly. To address this limitation, the present study adopted a strategy that integrates a combined feature-vector to ascertain the bag's overall classification. This approach diminishes the influence of any single instance's feature-vector, relying instead on the aggregate feature-vector for class determination. Consequently, this significantly reduces the likelihood of the entire bag being erroneously categorized as positive due to a solitary false positive instance. Recognizing the limitations of the conventional MIL assumption, introducing the concept of a combined feature-vector to predict the bag's label. This approach utilizes a deep Convolutional Neural Network (CNN) model with four input channels, simultaneously processing images from the same patient at various magnification levels. These images are categorized into binary groups Several preprocessing techniques are applied to improve the quality of histopathological images for more accurate cancer type prediction. Additionally, real-time data augmentation is incorporated into the process, and the feature learning backbone network chosen is EfficientNetV2-S.

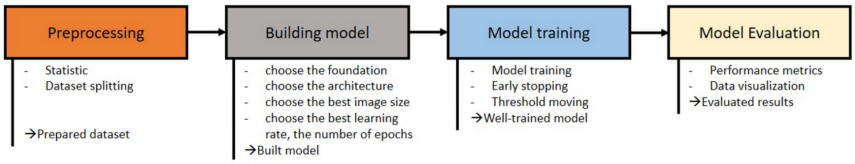


Fig. 1: Experimental Pipeline.

3.1 Dataset Description:

In this study, we have access to a variety of datasets containing histopathological images, which are integral for tumor tissue identification utilizing deep learning approaches. Specifically, the BreakHis dataset has been chosen for our analysis, having received the necessary permissions and acquisition of the dataset for

our research purposes. Our study incorporates the BreakHis dataset, which is composed of 7909 microscopic images depicting breast tumor tissues. These images, sourced from 82 different patients, are captured at varied magnifications including 40X, 100X, 200X, and 400X. The dataset, curated by P & D Lab in Brazil [21], is organized into two primary categories. It presents histopathological images classified under two distinct categories: benign and malignant, as illustrated in Fig. 2. A benign designation implies no tumor tissue presence, whereas a malignant designation confirms its existence. In the context of benign pathology, the classification delineates four distinct subcategories: Adenosis (A), Fibroadenoma (F), Phyllodes Tumor (PT), and Tubular Adenoma (TA) were consolidated into a single 'benign' category. Conversely, malignant pathologies are categorized into four subtypes, namely Ductal Carcinoma (DC), Lobular Carcinoma (LC), Mucinous Carcinoma (MC), and Papillary Carcinoma (PC) were amalgamated into the 'malignant' category. This reclassification resulted in the formation of two binary categories, with a total of 2,480 images classified as benign and 5,429 images classified as malignant.

Table 1: BreakHis Dataset.

Magnification	Benign	Malignant	Total
40X	652	1370	1995
100X	644	1437	2081
200X	623	1390	2013
400X	588	1232	1820
Total of images	2480	5429	7909

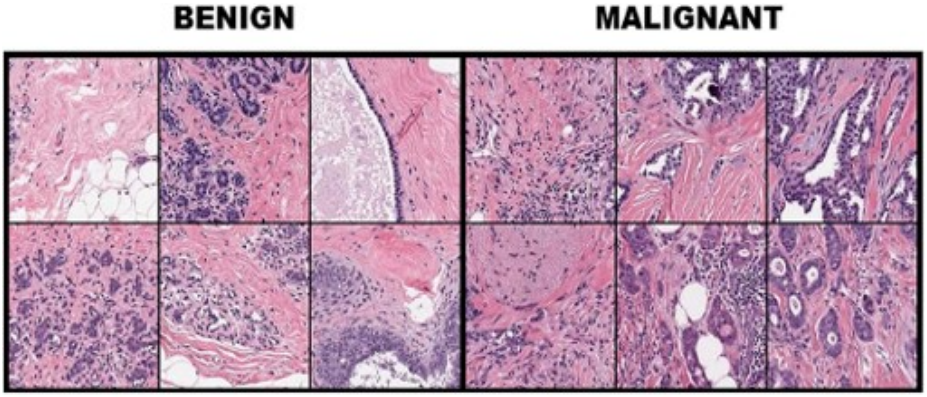


Fig. 2: Example of images available in BreaKHis Dataset.

3.2 Data Preprocessing:

Image preprocessing encompasses a series of techniques aimed at enhancing image data quality and accentuating critical features for later analysis, as indicated in source [20]. In the analysis of histopathological images, several preprocessing methods are applied to improve image definition and augment the accuracy of predictive models. The process began with the resizing of RGB images to 128×128 pixels, aligning with the requirements of EfficientNetV2-S, though this step modified the original aspect ratio. Subsequently, a median filter with a 3×3 kernel size was employed to minimize random noise while maintaining essential edge details, vital for later procedures like edge detection. As a nonlinear technique in image processing, the median filter enhances the image by updating each pixel's value to the median of the surrounding pixel values, thereby effectively reducing noise. To boost color depth, the images underwent a conversion from RGB to the HSV color space, followed by histogram equalization on the V (Value) channel, and then conversion back to RGB. Reflecting typical characteristics of medical data collection, the BreakHis dataset demonstrates common irregularities observed in medical datasets. A prominent feature of this dataset is the imbalance in data at various levels, particularly between malignant and benign samples at the image and patient levels, with respective ratios of 0.45 and 0.41. Furthermore, there is a marked disparity in the distribution of different subtypes at these levels, which may influence the efficacy of a Computer-Aided Diagnosis (CAD) system in binary and multi-classification tasks, favoring the more represented category. Although medical datasets often display imbalances, this study chose not to directly rectify class-level imbalances. Instead, we utilized a suite of evaluation metrics such as Matthew's Correlation Coefficient (MCC) [3] designed to address the data imbalance challenge.

3.3 Proposed Approach:

In this study, a sophisticated Convolutional Neural Network (CNN) with four distinct input paths was developed, tailored for processing histopathological images obtained at four different magnification levels. The images from each magnification level were fed into individual pathways of the CNN. The backbone of each channel was the EfficientNetV2-S model, a part of the advanced EfficientNetV2 series known for its enhanced performance in feature extraction and classification tasks. The EfficientNetV2-S model has a significant FLOPS count of approximately 8.8 billion. Within the EfficientNetV2 family [23], models vary in size and computational demand. The V2-S is an intermediate option, providing a balanced trade-off between performance and efficiency. It is more computationally intensive than smaller models like V2-Smaller or V2-Medium but less demanding than the larger V2-L. EfficientNetV2-S features a series of MBConv blocks (Mobile Inverted Bottleneck Convolutions) and Fused-MBConv blocks [23, 24]. These blocks are optimized for efficiency and performance, with a focus on reducing computational complexity while maintaining high accuracy. The model employs advanced activation functions like SiLU (Sigmoid Linear

Unit), also known as Swish [14]. This function is known for providing better performance compared to traditional ReLU, especially in deeper models like EfficientNetV2-S. EfficientNetV2-S offers a compelling balance between computational demand and model performance, making it ideal for tasks requiring both high accuracy and efficiency. Given its balanced design, V2-S is versatile and can handle a range of tasks, from feature extraction to complex classification, efficiently. After intrinsic feature extraction using EfficientNetV2-S, a feature vector of size 1×1280 is generated for each image at the four magnification levels: 40x, 100x, 200x, and 400x. The combined feature vector after concatenation from each group $\mathbf{A}_i = \{\mathbf{A}_{40i}, \mathbf{A}_{100i}, \mathbf{A}_{200i}, \mathbf{A}_{400i}\}$ is of size 1×5120, where $\mathbf{A}_{40i}, \mathbf{A}_{100i}, \mathbf{A}_{200i}, \mathbf{A}_{400i}$ be the feature vectors of the images after intrinsic feature extraction. This combined feature vector is then input into a fully connected layer consisting of four dense layers with 256, 128, and 64 neurons in the first three layers. The final layer's neuron count is adjustable based on the classification requirements, with a sigmoid activation typically employed for binary classification. The adoption of EfficientNetV2-S in this CNN architecture leverages its computational efficiency and advanced feature extraction capabilities, making it particularly suitable for the nuanced task of classifying histopathological images across multiple magnification levels. The architecture of the proposed CNN model is depicted in Fig. 3

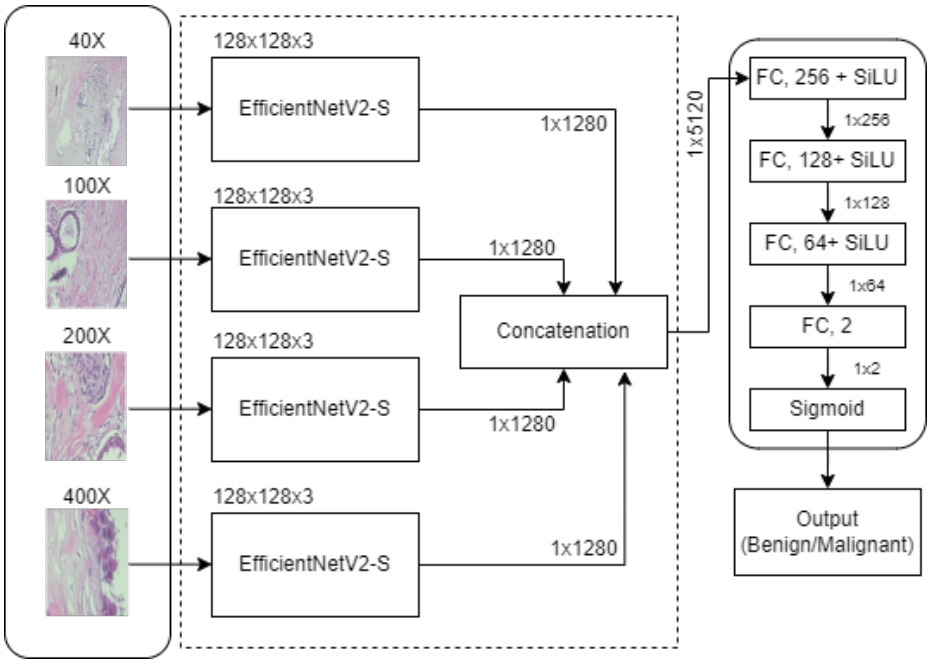


Fig. 3: Model Architecture.

3.4 Experimental setup and Hyper-parameter tuning:

For this study, the dataset was partitioned into training, validation, and testing subsets for each category independently. The test set constituted 20% of the total data. The remaining 80% was further divided, adhering to a training-validation split of 80:20. In alignment with the Multiple Instance Learning (MIL) framework, bags were constructed, each encompassing four images at varying magnification levels (40x, 100x, 200x, and 400x). Hyperparameter optimization was conducted via empirical trials. The learning rate was maintained at 1×10^{-4} throughout the training phase. Model performance was evaluated based on the validation loss, with the optimal model being saved when peak validation accuracy was achieved. Training was executed over 60 epochs, each comprising batches of 16 image bags. Binary cross-entropy served as the loss metric for categorizing images as benign or malignant. To mitigate overfitting, a dropout strategy was employed, randomly deactivating units with a 50% probability. Lion was chosen as the optimizer in this study. The experiments utilized Keras and TensorFlow frameworks and were executed on Google Colab Notebooks, offering a computational setup of 2vCPU @ 2.2GHz, 166.8 GB disk space, 51 GB RAM, and an NVIDIA T4 GPU with 16 GB memory.

Table 2: Hyperparameters utilized for training

Hyperparameter	Value
Total training epochs	60
Learning rate	0.0001
Batch size	16
Optimizer	Lion

3.5 Performance evaluation criteria

The classification efficacy was appraised employing an array of statistical metrics: the confusion matrix, accuracy is calculated by Equation 1, precision is given by Equation 2, recall as described in Equation 3, F1-score is formulated in Equation 4, Matthews correlation coefficient (MCC) is shown in Equation 5, and the area under the receiver operating characteristic curve (AUC). The confusion matrix elucidates the accuracy of the classification by displaying True Positives (TP) and True Negatives (TN) as indices of correct predictions, in contrast to False Positives (FP) and False Negatives (FN), which indicate inaccuracies. Accuracy is quantified as the proportion of true results (both true positives and true negatives) among the total number of cases examined. Precision is calculated as the proportion of true positives among the total positives predicted by the classifier. Recall, also known as sensitivity, measures the proportion of actual positives correctly identified. The F1-score is determined as the weighted

average of precision and recall, thereby balancing the two metrics. The AUC represents a summarizing metric, gauging the model's ability to discriminate across all possible thresholds. The Matthews Correlation Coefficient (MCC) is considered to be a balanced measure which can be used even when classes are of very different sizes, making it particularly informative for evaluating the performance of binary classifications on imbalanced datasets. The MCC takes into account true and false positives and negatives, which is advantageous because it provides a more insightful reflection of the model's ability to correctly identify positive and negative classes than accuracy alone, especially when the class distribution is imbalanced. MCC returns a value between -1 and +1, where +1 represents a perfect prediction, 0 is equivalent to random guessing, and -1 indicates total disagreement between prediction and observation. An MCC value close to +1 suggests that the classifier is performing well on both the majority and minority classes, whereas a value close to 0 would suggest it is performing no better than random chance, regardless of the imbalance [3]. These indices collectively form a robust framework for evaluating the predictive strength of the classification model. The metrics are formulated as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{4}$$

$$\text{MCC} = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{5}$$

4 Result, Discussion

Subsequent to the partitioning of the dataset, sets of instances, referred to as 'bags', were created. Each bag represents a sample comprised four images, each captured at a distinct level of magnification. These bags were assembled for the training, validation, and test datasets. The model's training was conducted utilizing the training and validation sets. Post-training, the model's efficacy was assessed based on its ability to classify the test set bags as either malignant or benign. The performance of the model during training and validation phases is illustrated through accuracy and loss curves for binary classification, as shown in Fig. 4. The binary classification outcomes are further encapsulated in a confusion matrix, presented in Fig. 5a. An analysis of this matrix reveals that the model achieved accurate classification of 209 malignant cases and 94 benign cases. However, discrepancies were noted where 6 benign samples were erroneously classified as malignant, and 3 malignant samples were misclassified as benign. It is pertinent to note that each sample within a bag encompasses four images, correlating

to four distinct magnification levels. Fig. 5b presents the Receiver Operating Characteristic (ROC) curve, plotting the true positive rate against the false positive rate across varying decision thresholds. The Area Under the Curve (AUC) was calculated to be 0.9978, indicating a high degree of discriminative performance by the proposed algorithm across all threshold levels. An MCC of 93.35% indicates a high degree of correlation between the observed and predicted classifications. In practical terms, this suggests that the model's predictions are highly congruent with the actual data. This level of MCC is remarkable, especially considering the challenges posed by the imbalanced nature of the dataset, where traditional metrics might overestimate the model's performance. The high MCC value underscores the model's effectiveness in distinguishing between malignant and benign cases, affirming its reliability in clinical or diagnostic applications where the cost of misclassification can be significant. In the present section,

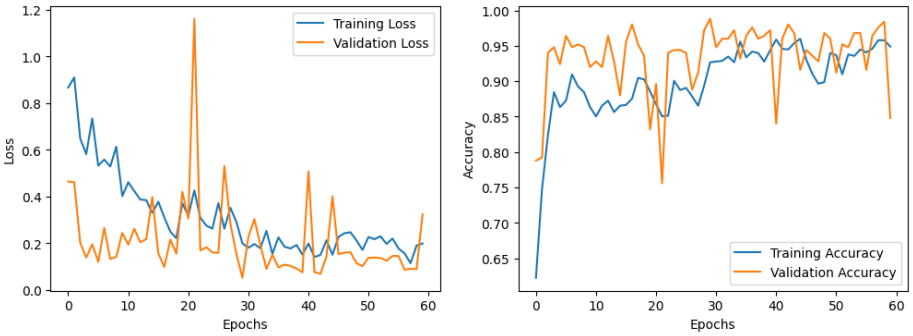
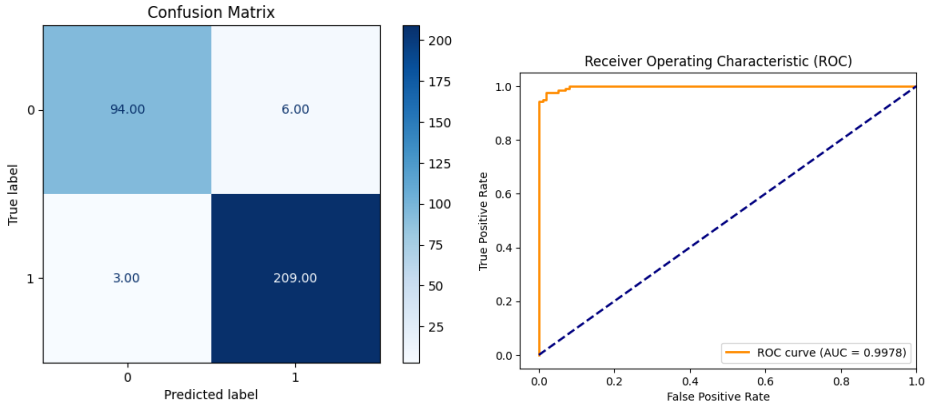


Fig. 4: Graph depicting the accuracy and loss values throughout the training and validation phases of binary classification. The epochs are shown on the horizontal axis, while accuracy and loss are represented on the vertical axis.

a comparative analysis is conducted with contemporary studies in this domain. The methodology introduced in this research demonstrates superior performance in four distinct classification tasks, surpassing current state-of-the-art methods. Specifically, in binary classification tasks, the model attained notable metrics: an accuracy of 97.12%, precision of 97.21%, recall of 98.59%, an F1-score of 97.89%, Matthew's Correlation Coefficient (MCC) of 93.35%, and an Area Under the Curve (AUC) of 0.9978. Referencing Table 3, it is evident that the proposed model yields enhanced classification efficacy compared to the works of using only transfer learning and classifying each magnifications such as Han et al. [10], Jitendra et al. [13], Prakash et al. [6], our approach also outperforms the application of traditional MIL and CNN of Kausik et al. [7] and attention-based MIL of Patil et al. [16].

The innovative Multiple Magnifications Learning approach, which processes multiple image magnifications to assign a singular label to a sample as opposed



(a) (b)

Fig. 5: Confusion Matrix (a) ROC curve and AUC score results (b).

to individual images, is a key factor in this enhanced performance. This approach leverages a combined feature-vector technique for class-label determination of the sample (referred to as a 'bag' in this context), deviating from the standard Multiple Instance Learning (MIL) assumption [5]. Under the standard MIL framework, a single false positive could lead to the misclassification of the entire bag. However, by employing a combined feature-vector approach, the influence of a single instance on the overall classification is significantly reduced. Consequently, the likelihood of erroneously classifying the entire bag as positive based on a single false positive instance is substantially diminished. This methodological refinement plays a crucial role in the observed performance metrics of the model.

Table 3: Comparison of obtained results with other state-of-the-art works on the BreakHis dataset.

Model	Accuracy	Precision	Recall	F1-score
CSDCNN [10]	93.20%	-	-	-
VGG16, ResNet [13]	90.40%	91.00%	90.00%	90.00%
Eff-net B2 [6]	92.15%	-	-	90.00%
MIL, CNN [7]	88.77%	-	-	-
A-MIL [16]	85.10%	-	-	-
Our approach	**97.12%**	**97.21%**	**98.59%**	**97.89%**

Table 4: Additional Results of our work on the BreakHis dataset.

Model	Accuracy	Precision	Recall	F1-score	MCC	AUC
Our approach	97.12%	97.21%	98.59%	97.89%	93.35%	99.78%

5 Conclusion

This study introduces a novel learning approach tailored for histopathological image analysis: Multiple Magnifications Learning that accommodates variable magnification levels. We developed a model with a quad-input pathway, designed to yield a singular output for an entire sample, circumventing the necessity for majority voting mechanisms. This model diverges from the traditional Multiple Instance Learning (MIL) paradigm by employing an integrated feature-vector concept. This methodology preserves the inherent relational features among images from the same patient, taken at different magnifications, thereby enhancing the diagnostic integrity of the analysis.

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